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Lunar Clinopyroxene and Plagioclase: Surface Distribution and Composition

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Abstract—The *Clementine* UVVIS images and the spectral and chemical (mineral) characteristics of lunar soil samples previously measured by the Lunar Samples Characterization Consortium were used to map the plagioclase and clinopyroxene abundance in the lunar surface material. An excess of plagioclase was found in young highland craters (e.g., in the crater Tycho) and in their ray systems. For clinopyroxenes, analogous behavior was observed in mare craters (e.g., in the crater Aristarchus). The maps for the FeO and Al_2O_3 bulk contents and the contents of these oxides in plagioclase and clinopyroxene were estimated by the same technique. These maps were compared to each other and to the predicted distribution of the lunar regolith maturity. The regolith of highland ray systems (e.g., the Tycho crater system) is characterized not only by low maturity but also by peculiar iron and aluminum contents: the lower the soil maturity degree, the smaller the iron content and the greater the aluminum content. This is confirmed by the data for the lunar soil samples from the *Apollo 16* landing site. A cluster analysis of the "clinopyroxene content–maturity" and "plagioclase content–maturity" correlation diagrams allowed the mineral mapping of the lunar surface to be performed.

INTRODUCTION

Plagioclase, pyroxene, and olivine are the main rock-forming minerals of the lunar surface (Frondel, 1978); plagioclase prevails in highlands, while pyroxene and olivine predominate in mare regions. The investigation of the regional abundance of these minerals and characteristics of their composition with the mapping method is one of the most important objectives of the remote sensing of the Moon. Only the first steps have been made in the development of this approach. These steps have been based on use of the Moon imagery and on the results of spectral and chemical (mineral) studies of lunar soil samples. In particular, the technique suggested for the estimation of the FeO and TiO_2 abundance in the well-known papers by Lucey *et al.* (1995, 1998, 2000a, 2000b) was based on such data. The improvement of the mineral prognosis technique is valuable not only for the continued analysis of the images received from the *Clementine* spacecraft but also for processing the data from new space missions such as *SMART 1* (Foing *et al.*, 2003; Shkuratov *et al.*, 2003a, 2003c).

The spectral characteristics used for the mineralogical prognosis substantially depend on the lunar soil

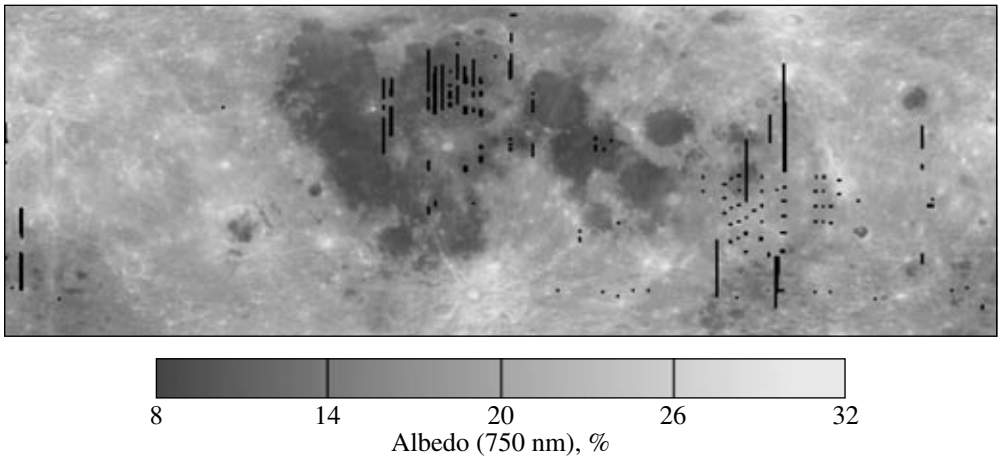


Fig. 1. The albedo distribution (0.75 μm) over the lunar surface from the *Clementine* UVVIS data. Vertical dark strips correspond to the areas not covered by the imaging.

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Table 1. The optical characteristics and the composition of the lunar samples examined in the study

Samples	415 nm	750 nm	900 nm	1000 nm	Al_2O_3, %	CaO, %	FeO, %	MgO, %	SiO_2, %	Al_2O_3 CPx, %	CaO CPx, %	FeO CPx, %	MgO CPx, %	SiO_2 CPx,%	Al_2O_3 Plag, %	CaO Plag, %	FeO Plag, %	MgO Plag, %	SiO_2 Plag, %
10084	4.61	8.00	9.04	9.62	16.2	12.53	12.22	7.34	47.5	18.07	9.32	11.19	0.79	3.16	5.59	3.00	6.77	2.78	8.25
10084	4.32	6.91	7.46	7.68	13.45	12.02	14.98	8.13	47.5	26.03	12.72	14.20	1.16	4.36	5.58	2.99	8.71	3.58	8.11
10084	4.38	6.45	6.40	6.29	12.19	11.78	15.74	8.59	46.5	31.92	17.16	22.63	1.46	5.95	5.52	2.98	7.23	2.35	7.82
12001	4.64	7.59	7.20	7.26	11.18	9.99	17.18	10.77	46.6	35.95	14.26	19.95	1.06	4.98	4.38	2.39	6.88	2.16	6.29
12030	9.91	15.44	13.51	13.64	10.83	9.76	17.41	9.98	47.1	38.03	13.62	19.48	0.93	4.71	4.53	2.44	7.43	1.96	6.60
12030	9.89	14.76	12.15	11.98	10.65	9.22	17.85	10.08	47.4	61.42	23.02	32.39	1.66	8.05	5.00	2.72	7.79	2.14	7.24
14141	13.58	20.15	16.03	17.35	15.00	10.10	11.60	11.00	45.7	21.30	18.48	19.58	1.69	6.11	8.89	4.67	1.59	2.39	12.13
14141	16.89	25.90	24.00	25.72	17.20	10.70	9.46	9.08	45.2	13.48	11.03	8.85	0.99	3.41	9.53	5.05	4.77	3.64	12.67
14163	6.09	10.22	9.42	10.07	15.40	10.20	11.50	11.00	46.2	16.46	16.07	13.36	1.35	4.90	6.20	3.32	1.13	0.75	8.71
14163	7.99	13.93	14.38	15.47	17.00	10.80	10.10	9.57	45.8	15.45	13.98	9.84	1.20	4.10	6.16	3.23	4.03	1.47	8.40
14259	4.87	8.09	8.21	8.77	15.80	10.50	11.00	10.70	46.3	21.50	17.42	14.30	1.57	5.41	4.74	2.45	0.71	1.41	6.55
14259	5.92	11.85	13.01	14.26	17.40	11.00	9.71	9.44	45.8	14.31	7.69	7.26	0.82	2.68	5.17	2.68	1.54	0.77	7.07
14260	4.63	7.94	7.90	8.53	16.30	10.70	10.70	10.40	45.6	18.90	14.94	11.34	1.32	4.51	5.26	2.75	2.81	1.41	7.12
14260	5.70	11.86	12.92	14.08	17.30	11.00	9.84	9.53	45.8	13.17	11.43	8.99	1.05	3.54	5.42	2.82	4.49	1.28	7.35
15041	4.18	7.63	7.91	8.33	13.70	10.35	14.61	10.96	47.1	30.74	9.71	14.67	0.76	3.62	5.38	2.87	6.98	2.92	7.65
15041	4.58	7.55	7.28	7.64	12.71	10.08	15.46	11.39	46.7	41.88	12.92	17.53	1.00	4.56	5.09	2.75	6.52	3.41	7.25
15071	4.73	8.78	8.81	9.25	13.07	10.34	15.60	11.15	46.5	30.89	10.37	16.02	0.77	3.84	6.51	3.56	7.19	3.11	9.03
15071	4.95	8.51	8.00	8.37	12.56	9.94	15.80	11.55	45.9	37.36	12.77	18.66	1.07	4.79	6.07	3.34	7.86	2.56	8.39
61141	7.59	12.70	13.51	14.51	26.10	15.20	5.15	6.56	44.3	3.44	5.34	2.56	0.40	1.37	14.85	8.04	4.25	1.70	18.85
61220	23.13	30.03	26.75	27.02	28.25	16.21	4.55	5.02	44.2	8.78	7.98	5.03	0.65	2.24	20.64	11.11	4.09	2.34	25.84
62231	10.39	14.95	15.93	16.86	25.70	15.30	5.31	6.59	43.7	4.23	5.78	3.40	0.44	1.54	14.13	7.79	6.06	2.83	17.64
64801	10.19	17.53	18.65	19.87	26.50	15.60	4.82	6.09	44.1	4.58	4.61	2.59	0.38	1.26	13.64	7.58	2.75	1.57	17.33
67461	19.43	28.13	27.76	29.42	27.30	16.10	4.93	5.11	44.0	6.61	8.21	4.97	0.62	2.18	22.52	12.36	9.65	3.86	28.31
67481	16.41	23.55	23.68	25.05	26.70	15.60	5.19	5.98	44.2	5.51	6.75	4.34	0.52	1.87	21.44	11.70	3.68	3.68	27.08
70181	5.57	10.20	11.83	12.79	15.68	11.71	12.93	9.29	45.2	8.29	3.50	4.57	0.28	1.21	6.23	3.39	2.96	2.22	8.36
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70181	4.73	7.06	7.23	7.44	11.73	10.51	16.32	10.31	45.7	27.98	14.53	16.46	1.32	4.95	5.59	3.04	5.07	2.54	7.72
71501	4.62	8.51	9.91	10.55	14.79	11.42	13.77	8.93	45.8	15.77	7.17	9.70	0.59	2.52	6.62	3.52	8.60	3.00	9.16
71501	4.60	7.55	8.11	8.33	11.85	10.32	16.76	9.73	45.9	24.68	11.13	14.92	0.96	3.96	6.65	3.52	7.33	2.97	9.09
71501	4.66	7.51	7.37	7.29	10.13	10.13	18.14	10.16	46.9	38.63	19.79	26.58	1.67	6.91	5.38	2.85	7.26	2.48	7.74
79221	5.20	9.06	10.45	11.36	16.22	11.94	11.53	9.78	45.4	6.15	2.10	3.47	0.20	0.84	6.23	3.33	6.51	3.53	8.44
79221	4.28	6.76	7.06	7.43	13.14	10.59	15.28	10.59	46.4	15.01	7.60	10.63	0.57	2.60	5.25	2.82	4.48	2.40	7.42
79221	4.68	6.89	6.94	7.15	11.86	10.53	16.16	11.15	45.6	22.32	11.25	17.89	0.87	4.06	5.61	3.06	5.41	3.21	7.71

composition and on the soil maturity. The maturity degree is characterized by the I_s/FeO parameter (Morris, 1976, 1977, 1978, 1980) that is a ratio of the abundance of reduced iron (this quantity is measured with the ferromagnetic-resonance method) to its total abundance. The maturity-degree mapping is a complicated problem; its solution has been outlined in several studies by Lucey *et al.* (2000b), Shkuratov *et al.* (2003a, 2003c), Kaydash *et al.* (2003, 2004), and Omelchenko *et al.* (2002).

Quite recently, Lucey and Steutel (2003), Shkuratov *et al.* (2002, 2003b, 2003c), and Omelchenko *et al.* (2003) published preliminary maps of the pyroxene abundance in the lunar soil. To map the pyroxenes, Lucey and Steutel (2003) used a model allowing the mineral composition to be calculated from the abundance of the main rock-forming oxides. Shkuratov *et al.* (2003a, 2003c) used the results of spectral and chemical (mineral) studies of several size fractions of lunar soil samples. These studies were carried out by the Lunar Soil Characterization Consortium (LSCC) (Taylor *et al.*, 2001a, 2001b). We found a correlation between the regolith spectral parameters and the pyroxene content and later used this dependence for calibration in predicting the pyroxene abundance throughout the lunar surface on the basis of the multispectral UVVIS imagery from the *Clementine* spacecraft (Shkuratov *et al.*, 2003b, 2003c).

The present study develops this approach. First, we extend it to the problem of predicting the plagioclase content. Second, we map the chemical composition of pyroxene and plagioclase. Another important distinguishing feature of the present study, as compared to our previous works, is the use of a new set of lunar samples, including both the mare and highland soils, delivered by the *Apollo 16* mission. The composition maps obtained in this study are analyzed together with the map of the lunar soil maturity degree, which was also built on the basis of the data for regolith samples.

INITIAL DATA AND MAPPING TECHNIQUE

To map the lunar surface in terms of chemistry and mineralogy, we used the results of the *Clementine* UVVIS imagery in five wavelengths: 415, 750, 900, 950, and 1000 nm (Nozette *et al.*, 1994). Initially, there was a large set of images with a spatial resolution of about 100 m. The images represented strips about 30 km wide under various illumination conditions. Then, the data for each wavelength were brought to a unified photometric system and were presented as a 1-km-resolution mosaic (McEwen *et al.*, 1998). Among the mosaics (maps), the 750-nm mosaic has the best photometric quality; its cylindrical projection is shown in Fig. 1. Dark vertical strips in the image correspond to the regions that were not covered by the imaging. For the purposes of mineral and chemical mapping, we used data for lunar soil samples that were obtained by the research consortium mentioned above (Taylor *et al.*, 2001a; 2001b). In Table 1, we give the information on 33 lunar soil samples for which the chemical and mineral compositions were measured (both the bulk composition and the contents of various minerals were determined). The spectral albedo in the UVVIS/*Clementine* bands and the data on the maturity degree are also presented in the table. It should be emphasized that the photometric systems used for the laboratory spectral measurements of the lunar soil and for the spectral zonal data received from the *Clementine* spacecraft are

Table 2. The correlation coefficients between the measured and predicted contents of the main minerals and the maturity degree I_s/FeO of the lunar surface

Mineral	Correlation coefficient
Plagioclase	0.93
Ilmenite	0.65
Olivine	0.49
Clinopyroxene	0.93
Orthopyroxene	0.54
I_s/FeO	0.88

Table 3. The parameters of Eq. (1) for clinopyroxene, plagioclase, and the maturity degree I_s/FeO

	a	*b*	*c*	*d*	*e*
CPx	−0.129	0.244	0.147	−0.321	1.27
Plag	0.114	−0.126	−0.021	0.093	1.064
I_s/FeO	0.006	−0.224	0.138	0.06	1.944

Table 4. The correlation coefficients for the measured and predicted contents of chemical elements in clinopyroxene and plagioclase

	CPx	Plag	Total
FeO	0.92	0.41	0.89
CaO	0.76	0.93	0.9
MgO	0.68	0.36	0.89
Al_2O_3	0.91	0.94	0.86
SiO_2	0.83	0.93	0.71

Table 5. The parameters of Eq. (1) for predicting the iron and aluminum contents in clinopyroxene

	a	*b*	*c*	*d*	*e*
CPx. FeO	−0.096	0.237	0.015	−0.215	1.427
Plag. Al_2O_3	0.118	−0.133	−0.024	0.1	0.575
Total FeO	−0.063	0.102	0.028	−0.112	1.376
Total Al_2O_3	0.052	−0.082	−0.046	0.107	0.992

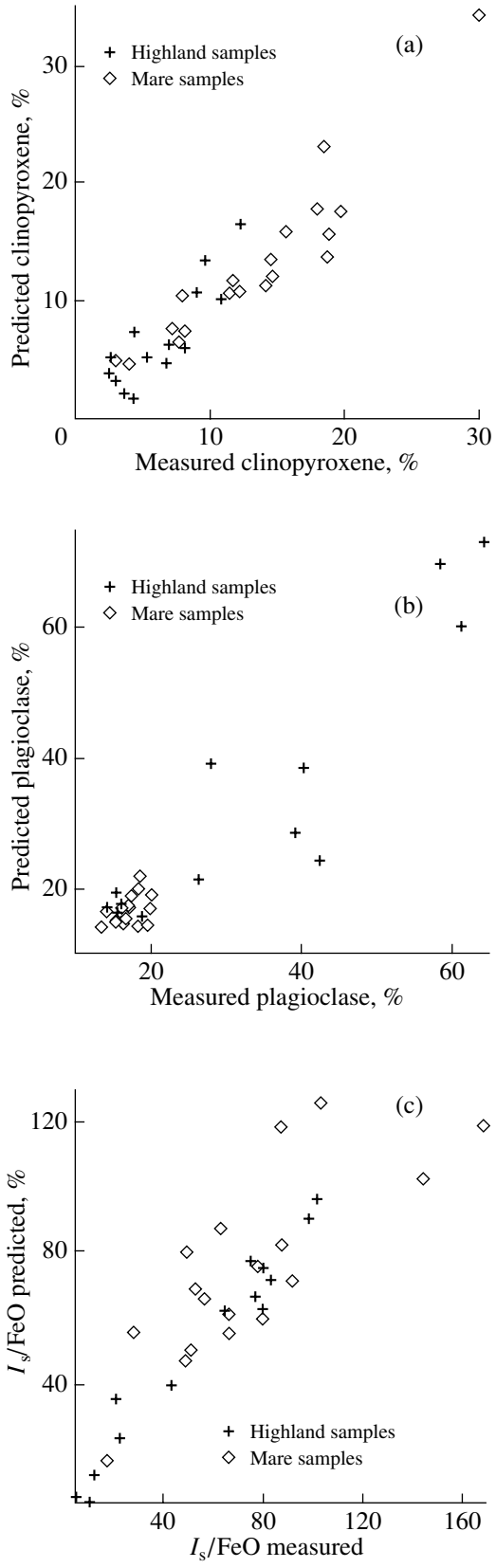


Fig. 2. The correlation diagrams for the measured and predicted contents of clinopyroxene (a), plagioclase (b), and maturity degree I_s/FeO (c).

the same. This is of principal importance for the analysis described.

A key feature of the present approach is the following. Among many combinations of optical parameters, we are searching for the one that most closely correlates with a chemical (mineral) parameter for a given set of lunar soil samples. We used a simple linear combination of the spectral albedo values in four wavelengths,

$$\log(P) = aA_{415} + bA_{750} + cA_{900} + dA_{1000} + e, \quad (1)$$

and obtained the maximal correlation coefficient by varying the constants a, b, c, and d. In order to avoid negative values of the predicted parameter, we used the logarithm of this parameter in (1). We ignored the 950-nm mosaic in our study because, first, it is close to the images obtained in the 900-nm and 1000-nm filters and, second, the correction and calibration of the 950-nm images are worse than those for images in other filters. It is worth noting that the choice of the main equation is optional to a marked extent. For example, this equation can be nonlinear. However, our analysis showed that various equations yielded close results (Omelchenko *et al.*, 2002; Shkuratov *et al.*, 2003a, 2003c). Therefore, the choice of simple linear dependence (1) seems to be well-founded.

An important question is whether the optical measurements in the UVVIS/*Clementine* spectral bands can be used to map the plagioclase and the abundance of nonchromophore elements (oxides), such as, for example, Al_2O_3. The spectral properties of regolith are determined by minerals—complex compounds of oxides—rather than by individual elements and oxides. The intensity and position of absorption bands are profoundly influenced by the distances between ions in minerals. Inserting nontransitional elements into the mineral crystal lattice changes the distances between the ions of transitional elements, which in turn affects the optical spectrum. A corresponding example is Ca in pyroxenes. Although Ca is not a chromophore element, it substantially changes the band positions of the pyroxene crystalline field near 1 and 2 μm (Adams, 1974). Note also that the content of the oxides and minerals that weakly influence reflectance properties sometimes closely correlates with the content of chromophore components; this produces correlations (sometimes rather high) between the albedo and the content of such nontransitional (optically neutral) elements as Si, Al, and Mg (Shkuratov *et al.*, 2004).

Table 2 presents the maximal correlation coefficients (from those that can, in principal, be obtained with Eq. (1)) for the measured and predicted values of the abundances of the main lunar-surface minerals and of their maturity degrees. Only clinopyroxene and plagioclase have high correlation coefficients (>0.9). The maturity degree also shows a high value of this coefficient (about 0.88). Further, we will analyze only these three characteristics, denoting clinopyroxene, plagioclase, and the maturity degree by CPx, Plag, and

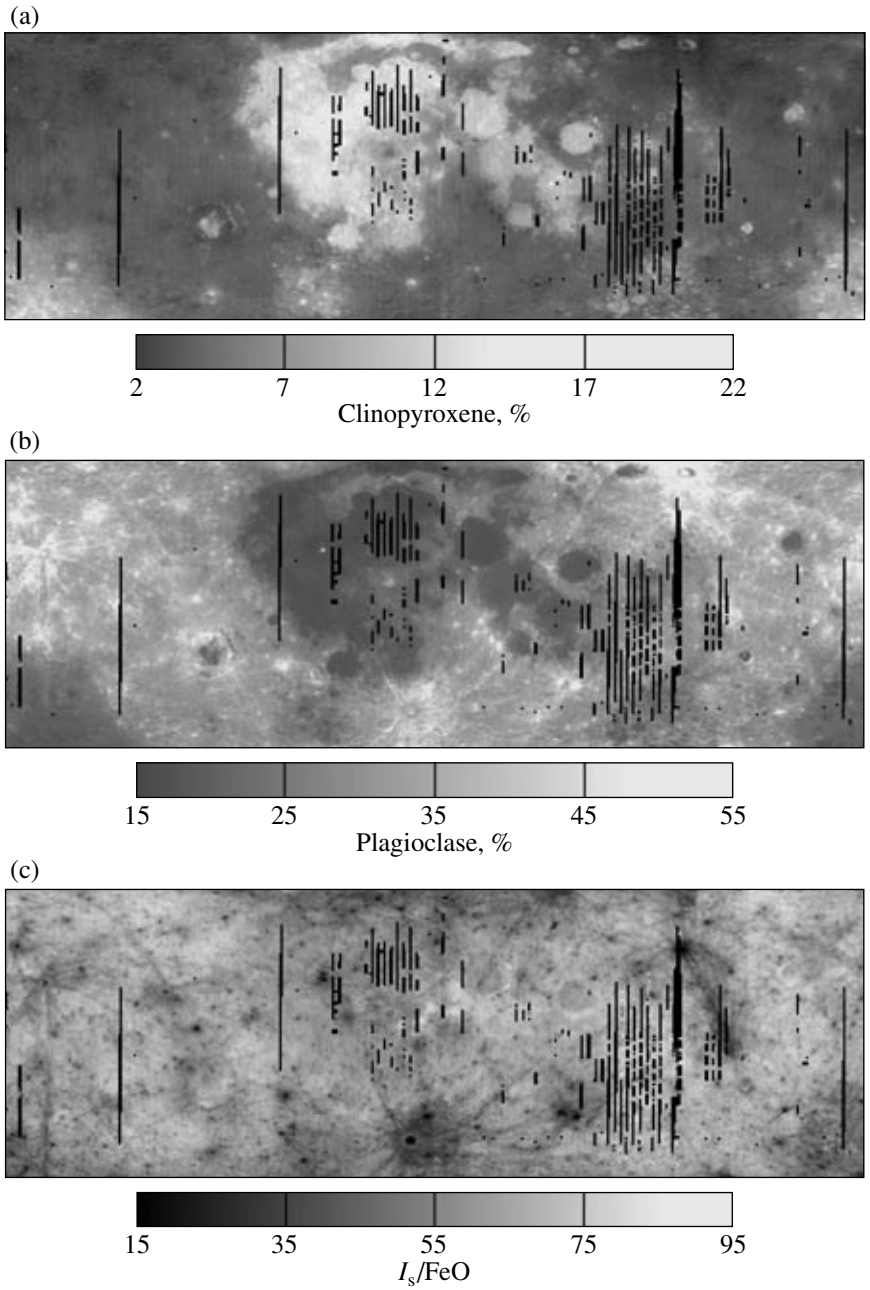


Fig. 3. The predicted distribution of clinopyroxene (a), plagioclase (b), and maturity degree I_s/FeO (c) over the lunar surface.

I_s/FeO, respectively. Table 3 lists the constants *a*, *b*, *c*, *d*, and *e* of Eq. (1) for the case when the albedo is expressed as a percentage. Figure 2 shows the predicted values of CPx (a), Plag (b), and I_s/FeO (c) versus the corresponding measured quantities. The correlation for plagioclase is seen to be produced by a small number of points at high values of its abundance (these points correspond to the highland material from the *Apollo 16* landing site). Nevertheless, this correlation is reliable: it is confirmed by the analysis of other sample sets.

In the same manner, we analyzed the bulk chemical composition and the composition of clinopyroxene and plagioclase of the lunar surface. Table 4 presents the coefficients of correlation between the measured and predicted values of the oxide abundances for some chemical elements. This table shows that the prognoses for the FeO oxide in CPx and the Al_2O_3 oxide in Plag are most reliable (the correlation coefficients exceed 0.9). These oxides were selected for mapping.

RESULTS AND DISCUSSION

Figure 3 displays the distribution of the CPx, Plag, and I_s/FeO parameters over the lunar surface. As might

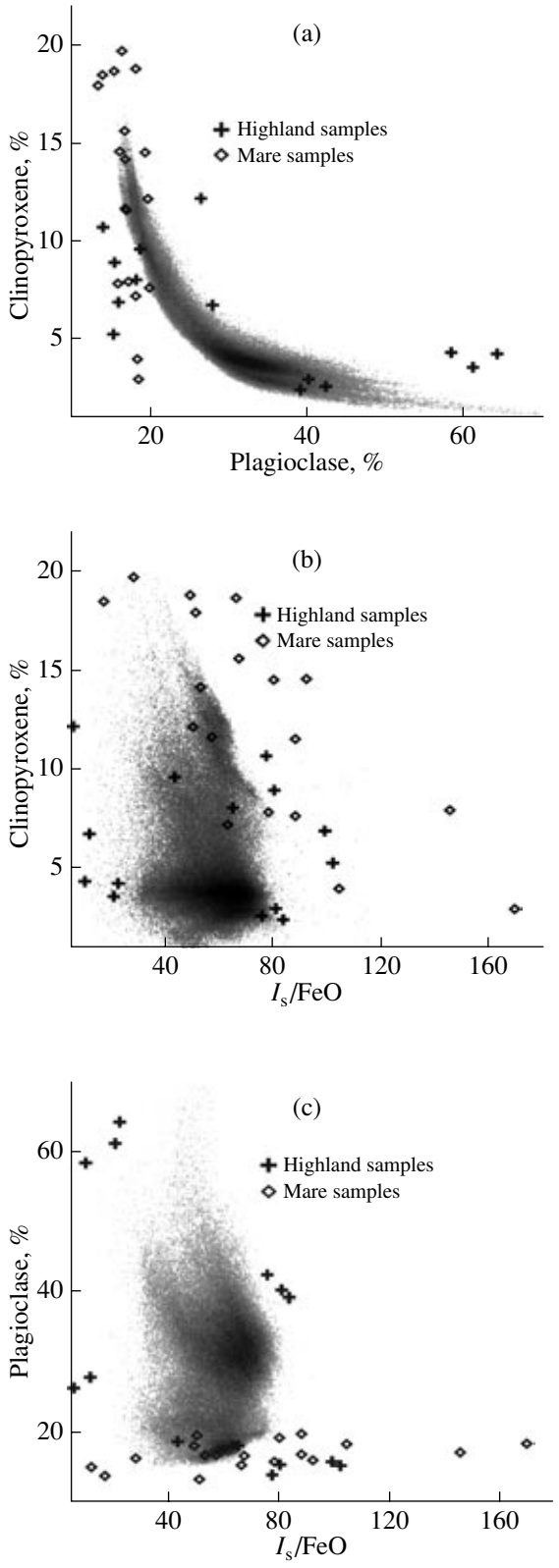


Fig. 4. The correlation diagrams for the contents of clinopyroxene and plagioclase (a), for the clinopyroxene content and maturity degree I_s/FeO (b), and for the plagioclase content and maturity degree I_s/FeO (c). The points represent the data for the lunar soil samples from Table 1.

be expected, the contents of clinopyroxene and plagioclase demonstrate an inverse correlation. The character of this dependence is seen in Fig. 4a, where the points corresponding to the lunar soil samples are shown as well. Note that the functions for the samples generally agree with the prognostic results, although the scatter in the data for the samples is substantially larger than that for the Moon. This can be naturally explained by a wider variety of samples (especially if we consider size fractions) as compared to the averaged soil characteristics of large surface areas.

The mineral abundance (a crystalline component) substantially depends on the surface age, which means that it must correlate with the soil maturity degree. Figures 4b and 4c present the correlation between the abundances of the examined minerals and the parameter I_s/FeO. It can be seen that both plagioclase and clinopyroxene depend on the maturity degree in a complex way, which is confirmed by the data for the lunar soil samples. In both cases, the mare and highland regions of the Moon correspond to different areas in the diagrams.

The CPx abundance map allows fresh craters to stand out prominently against the mare regions. A high CPx content in these areas (exceeding 20%) is due to a relatively low exposure age of the surface of the craters and their ejecta zones. In these areas the crystalline material, which also contains CPx, is not highly altered (vitrified) by micrometeoritic impacts. Consequently, there is more CPx there than in the other regions. The same can be said about the swirl Reiner Gamma, where some excess of CPx is observed in comparison with the surrounding mare regions. This agrees well with the age of the swirl: it is one of the youngest formations on the lunar surface (Pinet *et al.*, 2000; Starukhina and Shkuratov, 2004). The soil in the South Pole–Aitken basin (the lower corners of the mosaic) contains a lot of CPx. A high CPx content (>20%) is observed in the vicinity of the crater Aristarchus in Oceanus Procellarum. A reduced abundance of this mineral is observed in the areas of several fresh highland craters. This is probably related to the vertical heterogeneity of the lunar highland composition. However, a more conservative explanation would be the limitations of the technique applied. In actuality, there is no correlation for the points corresponding to the highland-soil samples in the CPx–I_s/FeO diagram (see Fig. 4b), while a weak inverse correlation is still observed for the mare soil.

The results of the cluster analysis of the CPx–I_s/FeO diagram are given in Fig. 5a. Four clusters have been distinguished (see the scheme in the right lower corner of Fig. 5a). Using the half-tone filling of the clusters, one can find the corresponding areas in the map (the darkest color in the map corresponds to the unclassified areas). As is seen in the cluster layout scheme, the CPx–I_s/FeO diagram has a fanlike structure: the extensions of all cluster axes approximately converge at the same point. An analogous structure was

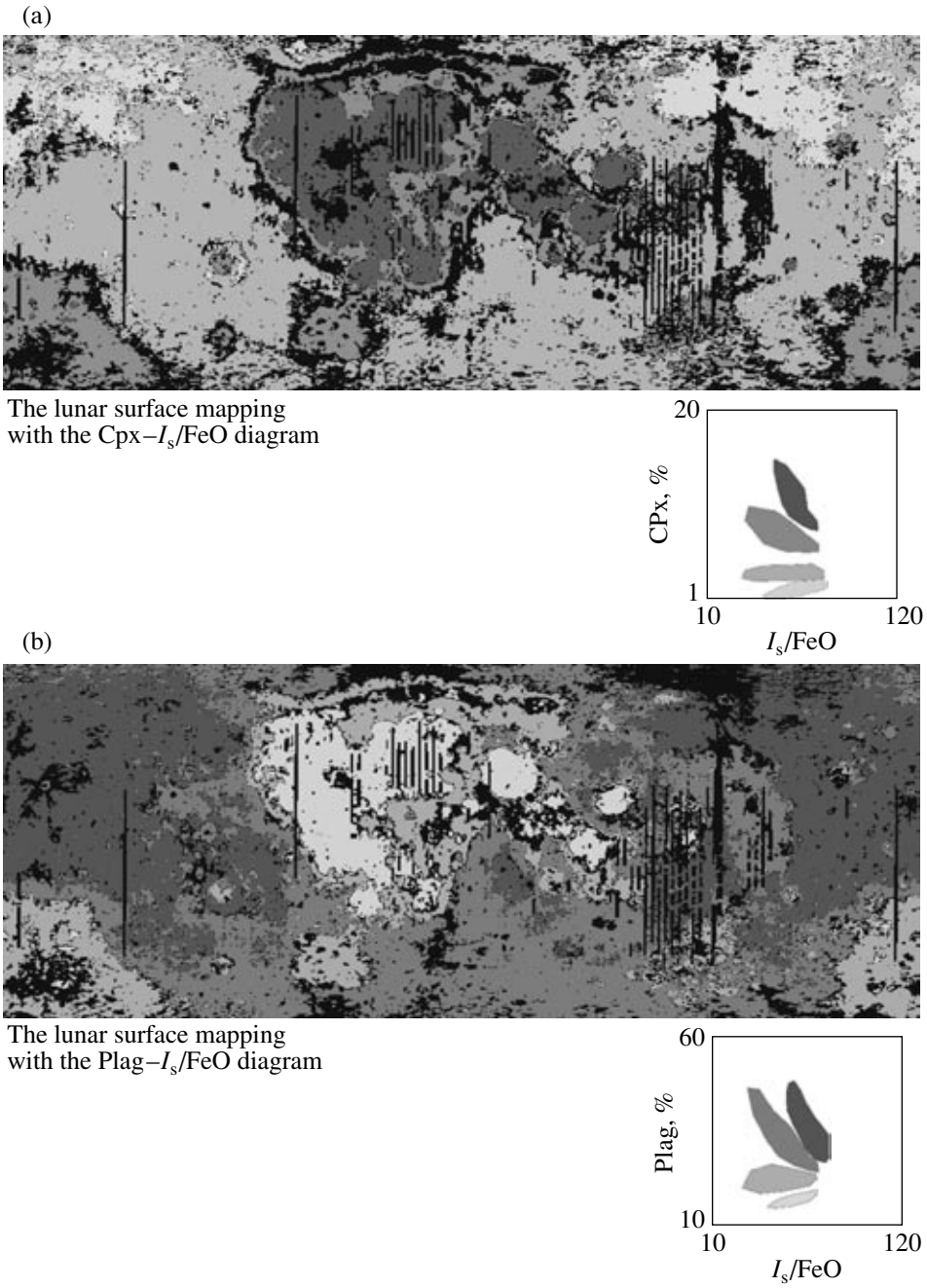


Fig. 5. The maps of mineralogical types of the lunar surface constructed on the basis of the CPx–I_s/FeO (a) and Plag–I_s/FeO (b) correlation diagrams.

found by Lucey *et al.* (1995) for the dependence of the color index A(750 nm)/A(900 nm) on albedo A(750 nm). Note that the shore region and a portion of the South Pole–Aitken basin are represented by a separate cluster.

A Plag excess (more than 50%) is observed in the highland craters; such an anomaly is not typical for mare fresh craters. The crater Aristarchus is an exception. During its formation, plagioclase was ejected by the impact from the highland bottom layer. A high plagioclase content in the material of fresh highland craters is explained in the same way as the CPx anomaly in the mare fresh craters: the plagioclase is better preserved in the places where the regolith has been less altered by meteorite impacts. The plagioclase abundance for Mare Tranquillitatis is somewhat higher than that for Mare Serenitatis. The lava flows along the western boundary of Oceanus Procellarum have a reduced plagioclase abundance. Wood's spot is also distinguished by a low content of plagioclase. A cluster analysis of the Plag–I_s/FeO correlation diagram was carried

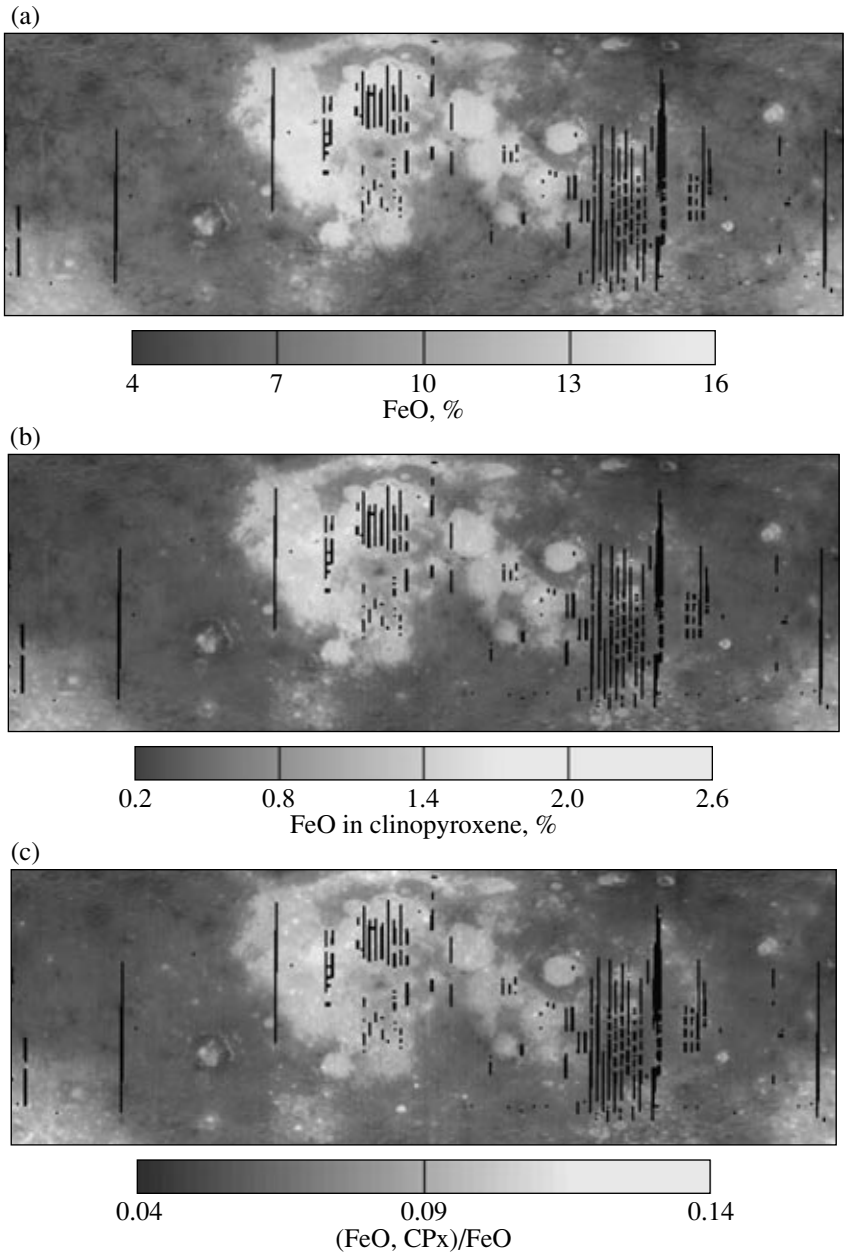


Fig. 6. The prognosis for the FeO oxide bulk content (a), the FeO oxide content in clinopyroxene (b), and the ratio of the FeO oxide content in clinopyroxene to its bulk content (c).

out (see the scheme in Fig. 5b). Four clusters are also distinguished here, as in the case of the CPx–I_s/FeO diagram. They are also arranged in a fan. The upper pair of clusters corresponds to the highlands of the lunar surface in the mineralogical map, while the lower one corresponds to the mare regions (the unclassified areas are black on the map).

A characteristic feature of the I_s/FeO distribution (see Fig. 3c) is a very weak mare–highland contrast; i.e., the maturities of mare and highland soils are almost the same. As expected, the maturity of fresh craters is low regardless of their mare or highland location. The maturity is extremely low near the Giordano Bruno crater and its ejecta. This is in good agreement with the fact that this crater is one of the youngest large impact formations on the Moon. The highest regolith maturity is observed in the regions that are classified by Wilhelms and McCauly (1971) as dark mantles. These areas are seen in the center of the image (light spots in Fig. 3c); the Sinus Meridiani can serve as an example.

We investigated the variations of the clinopyroxene and plagioclase compositions throughout the lunar surface. The oxides FeO in clinopyroxene and Al_2O_3 in plagioclase were mapped, and the maps were compared

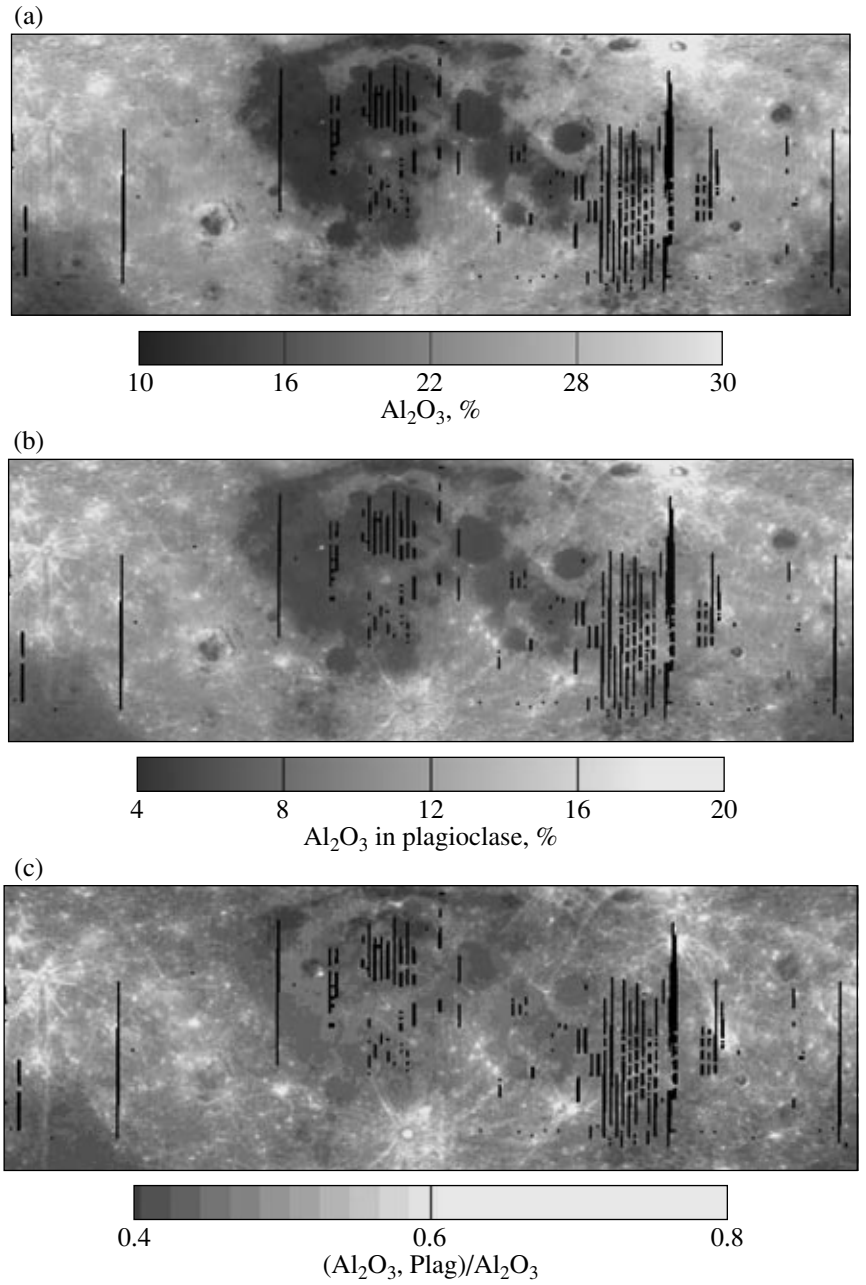


Fig. 7. The prognosis for the Al_2O_3 oxide bulk content (a), the Al_2O_3 oxide content in plagioclase (b), and the ratio of the Al_2O_3 oxide content in plagioclase to its bulk content (c).

to the bulk content distributions of these oxides (see Figs. 6 and 7). The distributions of the FeO bulk content and the FeO content in clinopyroxene differ from each other; this is demonstrated by the map of their ratio (Fig. 6c). The mare regions turn out to be homogeneous in FeO bulk content and inhomogeneous in the content of this oxide in clinopyroxenes, which is distinctly exhibited in the ratio map. A relatively high content of iron oxide in clinopyroxene compared to the mean bulk composition is observed for mare regions. Small-sized anomalies of this kind are also observed in the highland areas (see, e.g., the crater Tycho).

It is unusual that the FeO content correlates with the surface age. We used data on highland lunar soil samples (Morris, 1976, 1977, 1978, 1980) to find out whether such a correlation can exist in principle. Figure 8 displays the dependence of I_s/FeO on FeO constructed for the samples from the *Apollo 16* landing site; it definitely confirms the correlation. The origin of this correlation has not been analyzed before. It can be partly explained by the growth of the meteorite iron portion with increasing age of the surface (i.e., soil maturity). The other mechanism is the global contamination of highlands with mare material due to the surface mate-

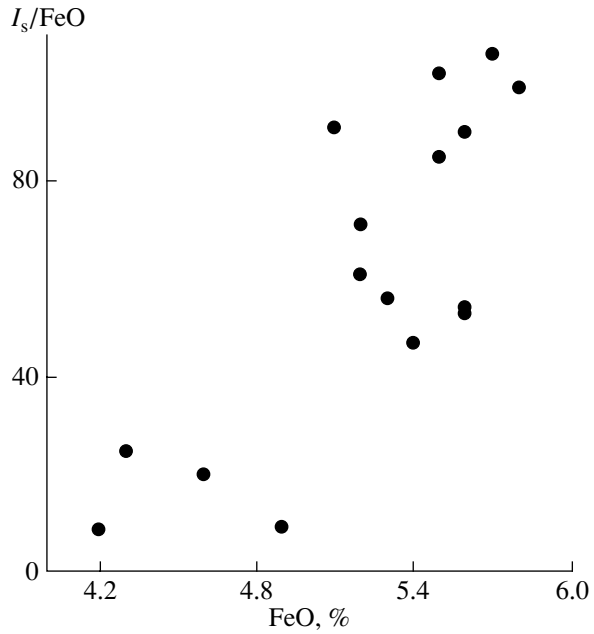


Fig. 8. The correlation between the value of I_s/FeO and the FeO content from the data for the soil samples from lunar highlands (Morris, 1976, 1978, 1977, 1980).

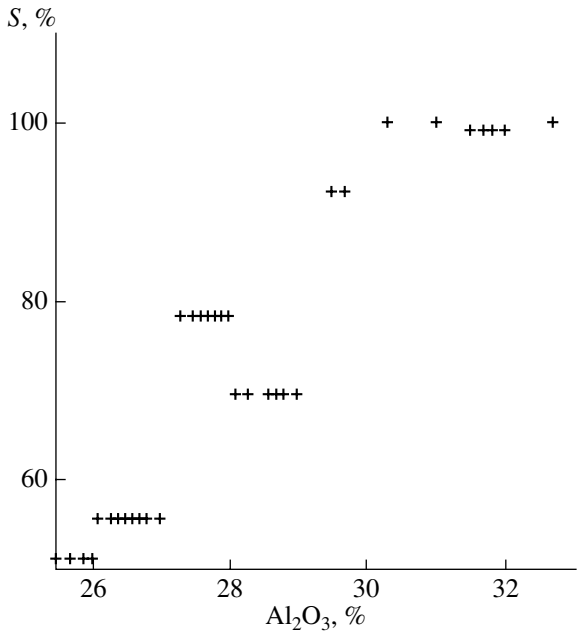


Fig. 9. The correlation between the contents of crystalline component, *S*, and the Al_2O_3 oxide for the soil samples from lunar highlands (Taylor *et al.*, 1991).

rial transport that accompanies meteorite impacts (Pieters and Taylor, 2003). The correlation can be also caused by the large-scale vertical heterogeneity of the lunar highland composition.

As in the case of FeO, the ray systems of fresh highland craters are clearly seen in the image of the Al_2O_3 bulk content distribution; in other words, the correlation between the maturity and composition, i.e., the Al_2O_3 content, is present again. We indirectly examined the possibility of such a correlation by using measurements of the *Apollo 16* samples. Unfortunately, sufficient data on the highland samples, which would include estimates of both the Al_2O_3 content and the I_s/FeO parameter, are lacking in the literature. Therefore, we used estimates of the content of the *S* crystalline component in the samples (Taylor *et al.*, 1991); the *S* parameter closely correlates with I_s/FeO. Figure 9 presents *S* as a function of Al_2O_3; the initial data (Taylor *et al.*, 1991) on the Al_2O_3 content scale were averaged in each 1% bin in order to smooth the curve. A clear correlation is observed, in agreement with the predicted composition maps. This correlation can be caused (as in the case of the FeO oxide distribution) by the global contamination of highlands with mare material (Pieters and Taylor, 2003) and by the large-scale vertical heterogeneity of the lunar highland composition.

The distributions of the Al_2O_3 oxide bulk content and its content in plagioclase are noticeably different. In the image of the ratio of the Al_2O_3 content in plagioclase to the bulk content (see Fig. 7c), the ray systems are well distinguished, and the heterogeneous features are clearly seen in maria. For example, the difference between Mare Traquillitatis and Mare Serenitatis is noticeable. This calls to mind the pattern seen in the image of the color index A(420 nm)/A(560 nm) of the lunar surface (Pieters, 1978).

CONCLUSIONS

At present, the mosaics of multispectral images of the Moon obtained by the *Clementine* UVVIS camera are unique in providing global coverage of the lunar surface by photometric measurements with a high spatial resolution in five spectral ranges. These measurements are reduced to the system of the lunar soil laboratory spectra measured with the RELAB instrument. For some lunar soil samples, not only optical characteristics but also data on their chemical (mineral) composition are available. Therefore, it is possible to simulate abundance maps for some lunar-surface oxides and minerals using the correlation between the spectral and chemical (mineral) characteristics of the lunar soil samples for calibration. The present study employs this particular approach to mapping the plagioclase and clinopyroxene contents as well as the soil maturity degree of the lunar surface material. The same technique is applied to build the prognosis map of the bulk content of FeO and Al_2O_3 and the content of these oxides in plagioclase and clinopyroxene. The listed characteristics—the content of plagioclase, clinopyroxene, FeO and Al_2O_3 oxides, and the maturity degree—were chosen because of the high correlation between the measured and predicted values of these parameters (for the samples examined). Other minerals and oxides, for

example, olivine, ilmenite, and CaO and MgO oxides, are predicted to be substantially worse; a study of them is beyond the scope of the present work, although there are some prospects in this field (Shkuratov *et al.*, 2004). Two specific results obtained in this study are indicated below.

1. A high content of plagioclase has been found in fresh highland craters and their ray systems. An analogous behavior is observed for clinopyroxenes in mare craters. The craters Tycho and Aristarchus can be considered as examples of fresh craters in highlands and maria, respectively. The above-mentioned anomaly can be caused by weak alteration of the crystalline lunar soil components (plagioclase and clinopyroxene) in young lunar surface areas.

2. Highland ray systems (e.g., the Tycho crater system) show not only low maturity but also peculiar iron and aluminum contents in their soil: the lower the soil maturity degree, the smaller the iron content and the greater the aluminum content. This is confirmed by the data on the lunar soil samples from the *Apollo 16* landing site. The revealed property can be caused by the global contamination of highlands with mare material due to impact events of various scales.

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